INCH-POUND

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10 July 1998
SUPERSEDING
MIL-E-1/1664A(NAVY)
16 June 1975

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 8827 *

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Coaxial, fixed frequency 16,410 to 16,625 MHz, pulsed, integral magnet, 50 kW rated peak power output.

ABSOLUTE RATINGS:

Parameter:	Ef	if (surge)	If	tk	tpc	rrv	Ib	ib	epy
Unit:	V	a	A	sec	µs	kV/µs	mA	a	kv
Maximum:	4	35	20	---	1.2	180	22	20	16.5
Minimum:	----	---	----	180	0.16	50	---	10	----

ABSOLUTE RATINGS:

				Temperature		Pressurization	
Parameter:	Pi	pi	VSWR	Anode	Input bushing	Input	Du
Unit:	W	kw	----	°C	°C	psia	----
Maximum:	275	275	1.5:1	175	200	25	0.0011
Minimum:	---	---	----	-65	-65	12	----

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Cooling:	Forced air 20/
Mounting position:	Any	Coupling:	
Mounting support:	Flange	Output:	Mates with UG-541
Cathode:	Pressurized 8/	Input:	Mates with Jettron P/N 90-001-C, or equivalent
Magnet isolation:	12/	Weight:	4.75 pounds maximum (2.154 kg)

TEST CONDITIONS: 3/

Parameter:	Ef	tpc	rrv	Du	epy	VSWR	T (anode)	T (bushing)	tk
Unit:	V	µs	kv/µs	----	kv	---	°C	°C	sec
Maximum:	2.84	0.55	170	0.001	16.5	1.1:1	165	190	---
Minimum:	2.57 2/	0.45 5/	6/ 7/	4/	15.5	---	-62 9/ 10/ 11/	-62	180

See footnotes at end of table I.

GENERAL:

Qualification: Required.

* Replaces Varian type No. BLM-199

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
Qualification inspection							
Salt spray (corrosion)	1006	14/	No voltages applied	----	----	----	----
Forced cooling	1143	11/ 19/ 20/		ΔT	----	75	°C
Temperature coefficient	4027	19/	T = 70° to 150°C	ΔF/ΔT	----	0.4	MHz/°C
Shock, specified pulse	1042	15/	Condition K; no voltages applied	----	----	----	----
High-frequency vibration	1031	15/	No voltages applied	----	----	----	----
High-temperature performance	1047	15/ 18/	No voltages applied	----	----	----	----
Low-temperature performance (1)	1047	15/ 16/	No voltages applied	----	----	----	----
(2)		15/ 17/	Snap-on	----	----	----	----
Frequency pushing figure	4311	21/	ib = 12 to 18 a	ΔF/Δi	----	0.5	MHz/a
Conformance inspection, part 1		22/					
Electrode current (anode)	1256			ib	5	13	a
Warmup time	4303		Ef = 3.5 V ac; tk = 150 sec (min)	If	10	11.5	A
Pulse characteristics	4304	5/ 6/ 7/		----	----	----	----
RF bandwidth	4308	23/		BW	----	1.32/tpc	MHz
Minor lobe ratio	4308	23/		Ratio	9	----	dB
Mechanical tuning range	4223		Upper limit	F	16,625	----	MHz
			Lower limit	F	----	16,410	MHz
Pulse voltage	4306			epy	15.5	16.5	kv
Stability	4315	24/		MP	----	0.5	%
Starting stability	4315	24/ 25/		MP	----	0.5	%
Power output	4250			Po	50	75	kW
Conformance inspection, part 2							
Frequency pulling figure	4310		VSWR = 1.5:1	ΔF	----	15	MHz

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
Conformance inspection, part 3							
Life test		1/ 5/ 26/	Group D	t	1,500	----	Hours
Life test end points:							
Power output	4250	13/		Po	40	----	kW
RF bandwidth	4308	13/ 21/		BW	----	2.5/tpc	MHz
Minor lobe ratio	4308	13/ 21/		Ratio	6	----	dB
Stability	4315	13/ 17/		MP	----	0.5	%

1/ Cycle life tests of 30 minutes “on” preceded by 3 minutes maximum filament schedule and 7 minutes “off” shall be required until the total accumulated “on” time equals the specified life. During the “on” periods, a VSWR of 1.5:1 maximum shall be introduced into external load circuit and the phase adjusted through 22 G during each “on” time.

2/ Prior to the application of high voltage, the cathode shall be heated to the required initial operating temperature. This may be done by applying 4.0 volts ± 5 percent for 3 minutes. On the application of anode power, the heater voltage should be lowered to the voltage specified. The tube heater shall be protected against arcing by the use of a connector that places a minimum capacitance of 4,000 pF across the heater directly at the input terminals.

3/ To relate the various parameters, use the following formula:

$$Pi = ib \times Du \times 16{,}000$$

4/ Heater voltage shall be 2.7 V for 0.0007 duty cycle.

5/ The characteristics of the applied pulse must be those which result in proper starting and oscillation. The rate of pulse voltage rise, the percentage of pulse voltage ripple, and the rate of pulse voltage fall are among the more important considerations. The cognizant service electron tube group should be consulted regarding pulse characteristics as related to the specific application.

6/ The rrv shall be expressed in kv for each μs defined by the steepest tangent to the leading edge of the voltage pulse about 80 percent amplitude. Any capacitance used in viewing system shall not exceed 6.0 pF. No spike or ripple shall exceed ± 7 percent of the average peak value of voltage or current. Inverse voltage shall not exceed 20 percent of the forward voltage. The negative amplitude of any post pulse voltage oscillation shall not exceed 10 percent of the average peak value.

7/ The rate of rise of voltage (rrv) shall be expressed in kv/μs defined by the steepest tangent to the leading edge of the voltage pulse above 50 percent amplitude. Any capacitance used in viewing shall not exceed 6.0 pF.

8/ The mounting plate shall provide for cathode terminal pressurization of 25 ± 2 psia and a leak rate not to exceed 0.1 standard cubic inch for each minute.

9/ With the specified cooling arrangements and conditions, the anode and input bushing temperatures shall not exceed the specified values.

10/ Unless otherwise specified, a total airflow of approximately 20 cfm shall be used for cooling.

11/ The temperature is to be measured at the points indicated on figure 1.

12/ In handling and mounting the magnetron, care shall be exercised to prevent demagnetization. A minimum distance of 6 inches shall be maintained between the tube and ferromagnetic materials or energized magnets.

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TABLE I. Testing and inspection - Continued.

13/ If during life test the tube does not meet the specified limits, it shall be recycled for an additional five cycles. At such time the tests shall be repeated. If the tube fails the second test, it shall be considered unsatisfactory.

14/ The interior of the waveguide flange, cathode terminal, and portion of the mounting plate adjacent to cathode terminal may be sealed off during this test to simulate the installed condition.

15/ After subjection to the specified test, the tube shall meet the requirements for rf bandwidth, minor lobe ratio, and stability as specified.

16/ The magnetron shall be stored for 72 hours at a chamber ambient temperature of -62°C (-80°F).

17/ The tube shall be allowed to thermally stabilize at -54°C (-65°F); apply standby filament schedule (3 minutes minimum) then turn magnetron on and off several times to assure “snap on” capabilities. Operate for 1 additional hour. The rf envelope shall be monitored to evaluate operation.

18/ The magnetron shall be stored at +85°C (+185°F). Thermally stabilize for 4 hours.

19/ Temperature measurements shall be made only after thermal equilibrium has been reached.

20/ A total airflow of 10 cfm at approximately 760 mmHg and 25°C, shall be directed into the fins. The specified rise above ambient temperature shall not be exceeded.

21/ The pushing factor shall be measured in three steps, 2 amperes each (10 to 12, 12 to 14, and 14 to 16), and no value shall exceed the limits specified herein.

22/ Unless otherwise specified, the acceptance level for all tests listed under conformance inspection, part 1, shall be zero, inspection level II.

23/ The magnetron shall be operated into a transmission line with a VSWR of 1.5:1 minimum adjusted in phase to produce the maximum spectrum degradation.

24/ Stability shall be measured in terms of the average number of output pulses missing, expressed as a percentage of the number of input pulses applied during the period of observation. The missing pulses (MP), due to any cause, are considered to be missing if the rf energy is less than 70 percent of the normal energy level. The VSWR shall be adjusted to that phase producing maximum instability and the MP counted during the last 3 minutes of a 5-minute test period.

25/ Immediately after a nonoperating holding period of 168 hours minimum, the starting stability test shall be performed on 100 percent of the submitted lot.

26/ Life test shall be conducted in accordance with the following cycle:

Condition	Ib	Ef	Duration
Standby	0	3.5	3 minutes
Test 1	13.5 ma	2.7	30 minutes
Off	0	0	7 minutes

27/ Revision letters are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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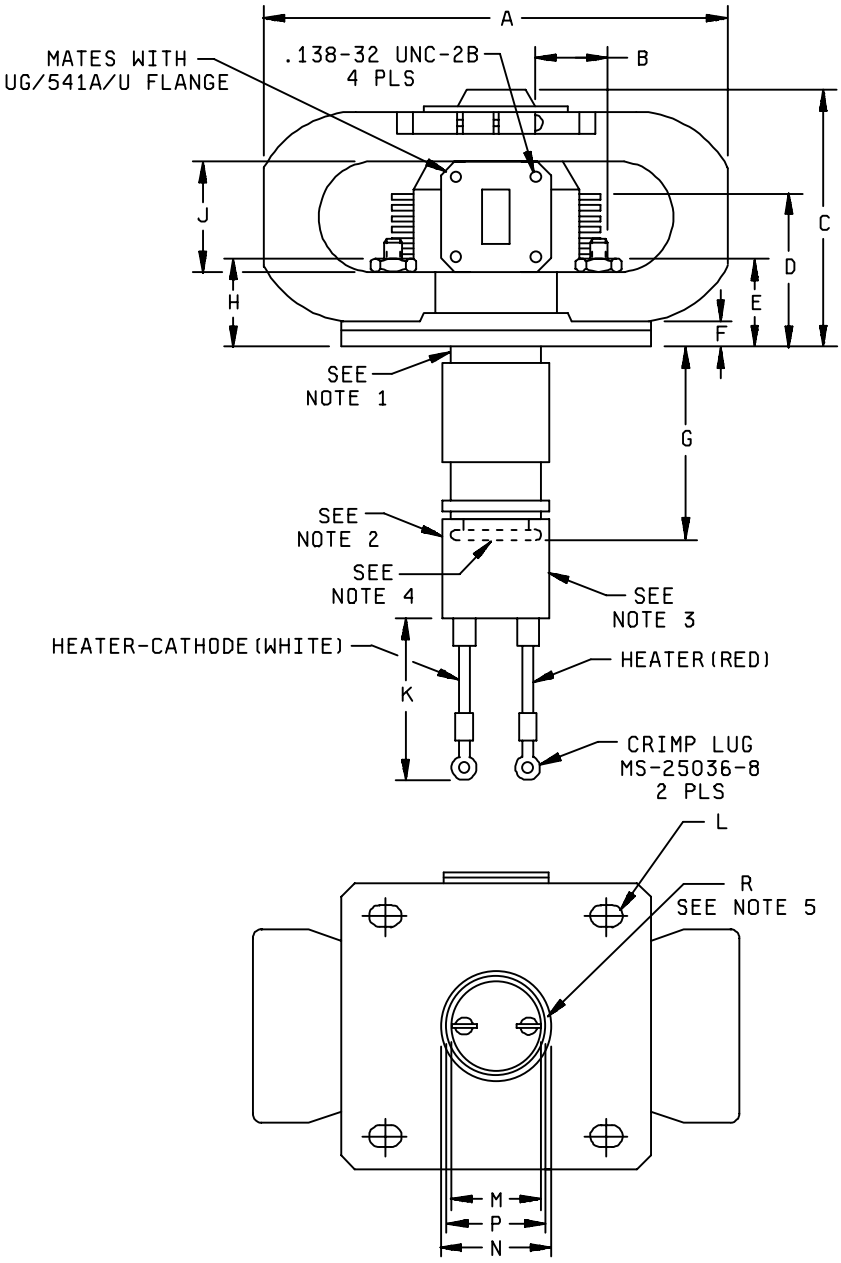

FIGURE 1. Outline drawing of electron tube type 8827.

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Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Qualification inspection				
A	----	5.800	----	147.32
C	----	3.000	----	76.20
G	----	2.190	----	55.63
M	----	1.100	----	27.94
Conformance inspection, part 1 (see 22/ of table I)				
B	0.866	0.926	22.00	23.52
D	----	1.922	----	48.82
E	0.995	----	25.27	----
F	----	0.250	----	6.35
L	0.129 R	0.141 R	3.28 R	3.58 R
Conformance inspection, part 2				
H	0.980	1.020	24.89	25.91
N	1.745	1.755	44.32	44.57
P	1.345	1.355	34.16	34.42
R	0.111	0.121	2.82	3.07
Nominal				
J	1.312		33.32	
K	2.312		58.72	

NOTES:

a. Anode temperature shall be measured at this point.

b. Bushing temperature shall be measured at this point.

c. Jettron Products, Inc. Connector No. 90-001, or equivalent.

d. Protective enclosure.

e. This dimension applies to depth of “O” ring groove.

FIGURE 1. Outline drawing of electron tube type 8827 - Continued.

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Custodians:
Navy - EC

Review activities:
Navy - AS, CG, MC, OS

Preparing activity:
DLA - CC

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